

Network Intrusion Detection Using Big Data Analytics – A Study on UNSW-NB15 Dataset

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Abstract—Intrusion detection systems (IDSs) are currently drawing a great amount of interest as a key part of system defense. IDSs collect network traffic information from various points on the network or computer system and then use this information to secure the network. Distinguishing between normal and malicious network activities is challenging and time-consuming due to the large volume of data and complex patterns involved. Traditionally, an analyst must review extensive datasets to identify sequences of intrusion within network connections. Thus, there is a need for efficient methods to detect network intrusions that reflect current network traffic conditions. In this study, a novel method is proposed to identify intrusion characteristics for IDSs using a stacking classifier in machine learning. The method involves generating rules through classification by a stacking classifier that combines multiple models to improve prediction accuracy. These rules can identify intrusion characteristics and are implemented in the stacking classifier to not only detect the presence of intrusions but also to execute preventive actions by denying intrusions. The classification of attack types includes: Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms, and Normal. The evaluation techniques used to assess the performance of the proposed method include Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). This approach aims to enhance the effectiveness of IDSs by leveraging the strengths of ensemble learning, thus providing a more robust and accurate detection and prevention mechanism for network intrusions.

Keywords: *Intrusion Detection System (IDS), Network Security, Machine Learning, Stacking Classifier, Ensemble Learning, Attack Classification.*

I. INTRODUCTION

Intrusion Detection Systems (IDSs) are essential for keeping computer networks secure and reliable. As internet use, cloud computing, and IoT devices continue to grow, networks face more complicated and advanced cyber-attacks. Traditional IDS methods that use rules or known attack signatures often fail to detect new or changing threats, making systems more vulnerable. This highlights the need for better detection tools that can handle the changing nature of network activity and

accurately tell the difference between normal and harmful behavior. Datasets like UNSW-NB15, which include many different types of attacks alongside regular traffic, offer a chance to build and test strong models that can reflect real network conditions.

In recent years, machine learning has become a useful tool for improving IDS performance. These models can automatically discover complex patterns and relationships in large network data, which is hard for humans to handle. However, using single models like Support Vector Machines, Decision Trees, or Neural Networks has its limits. These models might not work well on different types of attacks or may focus too much on certain traffic patterns. Also, imbalanced data, where one type of attack is much less common than normal traffic, can lead to unfair results and poor performance on rare attacks. This has led to more interest in ensemble learning methods, which combine multiple models to improve overall performance, making systems more reliable and adaptable.

Even though there's progress in using machine learning for intrusion detection, challenges remain in achieving high accuracy and reliability for all types of attacks. Many systems focus on overall accuracy but ignore important metrics like precision, recall, and F1-score for specific attack categories. This means some models work well for common attacks but miss rare but dangerous threats like Worms or Shellcode. Additionally, many IDS frameworks aren't built for real-time use, which makes them unsuitable for large, fast-moving network environments. These issues show that there's a need for a more complete and scalable solution that includes good data preparation, balanced evaluation metrics, and advanced ensemble strategies.

The goal of this work is to create and implement an efficient IDS using a stacking classifier, a type of ensemble learning that combines multiple base models with a meta-model to improve results. The system will use the UNSW-NB15 dataset to train and test models that can identify different intrusion types, including DoS, Exploits, Fuzzers, and others, along with normal traffic. By using preprocessing methods, thoughtful feature selection, and stratified evaluation, the system aims to

address class imbalance and improve the detection of less common attack types. In the end, this approach should not only increase detection accuracy but also offer a scalable and practical solution for modern networks, helping to better defend against ongoing cyber threats.

II. LITERATURE SURVEY

Network intrusion detection systems (IDS) have been on the center of attention of research because of the growing threats to cybersecurity. Such systems play a crucial role in detecting a network intrusion or unauthorized operations, and the systems have undergone development. Since the progress in machine learning (ML), deep learning (DL), and big data analytics, IDS became more precise and could process large amounts of data in a real-time. This literature review will explore the new developments in IDS especially in connection to using big data analytics and the UNSW-NB15 dataset, a popular network intrusion monitoring benchmark.

The first-generation intrusion detection systems mainly used signature-based mechanisms of intrusion detection, which had a low capability of detecting new attacks. With the introduction of machine learning, however, IDS systems started to shift to an anomaly-based detection model in which a model has been trained to identify abnormal patterns in network traffic. He [1] is an artificial intelligence lightweight IDS that has been suggested to minimize the amount of computation and enhance the accuracy of detection. Hossain and Hoque [2] proposed real-time IDS based on machine learning and Apache Kafka and emphasize the significance of real-time data processing in current IDS.

Modern IDS has focused on machine learning methods. Dietz et al. [3] addressed the application of AI/ML in enhancing the efficiency of network intrusion detection systems by bringing the research efforts to the users. This change to AI-based solutions has made it possible to detect more dynamic complex attack patterns. Qazi et al. [4] also investigated the concept of applying deep learning in IDS and suggested a Zero-Touch Network Security (ZTNS) that automates the whole process of detection and response, which proves the strength of the deep learning models in real-time network monitoring.

Ben Said, Triki, and Jemai [5] suggested a hybrid deep learning model based on CNN-BiLSTM trained to recognize intrusion in software-defined networking (SDN). Their work uses hybrid feature selection techniques in order to improve detection accuracy. To that effect, Sahu and Nayyar [6] designed a distributed big data-oriented intrusion detection scheme with the primary focus on scalability and distributed processing in the contemporary IDS. This model is applicable particularly in situations whereby extensive datasets are involved, e.g. UNSW-NB15 which has a sizeable concentration of tagged network traffic information.

Big data analytics is a very important tool in augmenting the performance of IDS especially when using the large scales of the dataset. Ahmad et al. [7] have shown how Apache Spark can be used in big data IDS to provide a scalable IDS system to handle a large amount of network traffic data. Wu, Chen, and Wang [8] suggested a better IDS, which integrates deep-learning with features optimization to improve the detection rate based on the UNSW-NB15 data. They propose their approach based on deep learning to extract the features that are relevant and enhance the performance of the detection system.

Kasongo and Sun [9] have examined the performance of various IDS models on the UNSW-NB15 dataset on feature selection methods, which is an important insight into the strengths and weaknesses of the various feature selection methods. Talukder et al. [10] applied the oversampling and feature embedding technique to address the imbalanced data in IDS and demonstrated that both strategies can enhance the efficiency of network intrusion detection systems on the UNSW-NB15 dataset.

The combinations and integration of several models or hybrid methods have been also discussed in the context of IDS. Yin et al. [11] offered an ensemble classifier model that is fine-tuned on the dataset features to detect network intrusion. Their work has highlighted the need to make the appropriate choice of models to get the best performance. More et al. [12] aimed at improving the performance of IDS through integrating various machine learning models and applying them to the UNSW-NB15 data. They showed how hybrid systems would be able to provide better detection rates than conventional ones would.

Ajagbe [13] made a comparison of the different machine learning models using the UNSW-NB15 dataset to demonstrate the superior performance of ensemble classifiers in terms of higher detection rates. Turk [14] also examined the application of machine learning algorithms to UNSW-NB15 and NSL-KDD datasets giving a detailed analysis and evaluation of the performance of the models on the various datasets.

Detecting imbalance and overlap of data is still a problem in intrusion detection. Yin et al. [15] aimed at minimizing the margin of error that occurred due to overlapping data and class imbalance in network traffic datasets. Their study presented a new hybrid feature selection mechanism to enhance the accuracy of detection. Zoghi and Serpen [16] also discussed the issue of the data imbalance in IDS, noting that it is essential to apply effective models that can address the problem.

Sajid et al. [17] proposed hybrid machine learning and deep learning-based intrusion detection that enhances its performance through mitigating the issue of class imbalance, as well as applying ML and DL methods. Their research shows

that a balanced approach to use is whereby strengths of various algorithms are integrated to enhance detection accuracy.

In a recent study, Mamatha et al. [20] indicated the creation of hybrid IDS based on using ensemble feature selectors and learning classifiers to curb Denial of Service (DoS) attacks. Their research highlights how there is increased pressure to use various levels of detection methods, which incorporate alternative feature selection schemes and classifiers to combat intricate threats to networks. With the still growing network traffic, the application of big data analytics and machine learning will play a central role in creating stronger and more scalable IDS systems.

To sum up, network intrusion detection systems have recently demonstrated considerable progress with the help of machine learning, deep learning, and big data analytics. The UNSW-NB15 dataset has been instrumental in the creation and testing of these systems, and offers researchers a large scale and real-world dataset with which to test intrusion detection techniques. Future studies will probably be aimed at enhancing the scalability, accuracy and effectiveness of IDS by taking additional advantage of big data technologies and hybrid machine learning models.

III. PROPOSED METHODOLOGY

The proposed approach describes how to create a strong Intrusion Detection System (IDS) using the UNSW-NB15 dataset, better data preparation methods, selecting the best features, and using stacking with multiple models. The process is broken down into different parts, as explained below.

A. Data Collection

The UNSW-NB15 dataset, made by the Australian Centre for Cyber Security, is used in this study. It has a mix of normal and attack traffic, with 49 features that describe the flow of data, content, timing, and other statistical details. The dataset includes nine types of attacks like DoS, Exploits, Reconnaissance, Backdoors, Worms, Shellcode, Generic, Analysis, and Fuzzers, along with normal traffic. Because of its variety, it is a good choice for testing IDS models in real-world situations.

B. Data Preprocessing

To make sure the data is consistent and reliable, several steps are taken:

Fixing Missing Data: Any incomplete or missing entries are filled with statistical methods like using the mean or median value.

Scaling Numbers: Continuous data points are adjusted using Min-Max scaling to bring them within a range of 0 to 1, which helps the model learn more effectively.

Converting Categorical Data: Features like protocols, services, and flags are turned into numbers using encoding techniques.

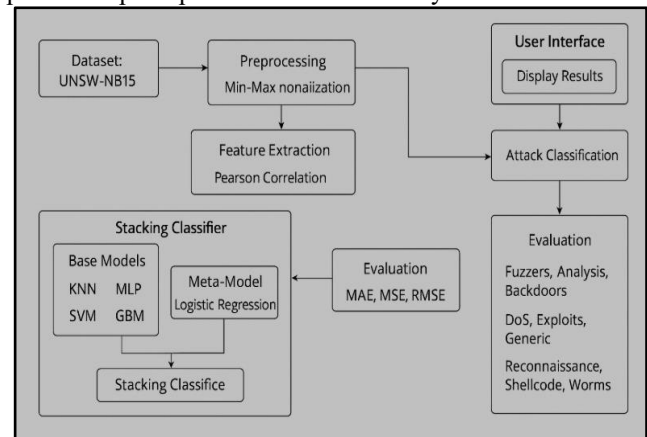
Splitting the Data: The dataset is shuffled and split into 80% for training and 20% for testing to evaluate the model fairly and avoid bias.

C. Feature Selection

High-dimensional data can be messy and slow for models to process. The Pearson Correlation Coefficient is used to measure how closely features are related to the target class. Features that are not very useful are removed, which makes the data simpler and more efficient without losing important information. This helps the model train faster and perform better.

D. System Architecture

The system's design, outlined in Figure 1, includes several important parts. To start, the UNSW-NB15 dataset is used as the data source, which includes both normal and malicious network traffic. Before processing, the data is cleaned up by fixing missing entries, adjusting feature values to a standard range, and converting categorical data into a format that's easier to work with. Then, a method called Pearson correlation is used to choose the most relevant features, which helps cut down on unnecessary information and makes the process more efficient. The cleaned and selected data is then passed into a stacking ensemble model. This model uses several different base models—like KNN, MLP, SVM, and GBM—to make predictions. These predictions are then combined using a Logistic Regression model that acts as a meta-classifier. This approach helps improve the model's ability to detect different



types of attacks accurately.

Fig. 1. System Architecture of the Proposed Intrusion Detection System

The classification part of the system provides results in two ways: one is a simple yes-or-no classification (normal vs. intrusion), and the other is more detailed, identifying the exact type of attack. Finally, the system is put into a real-world setup using tools like Flask or Fast API for the interface, Docker for containerization, and Nginx for handling web traffic. This setup allows the model to work in real-time, keep logs, and send

alerts. The overall design ensures the system is reliable, fast, and ready for practical use in detecting network intrusions.

E. Stacking Ensemble Learning

The main idea of this method is stacking, which is a type of ensemble learning.

Level-0 Base Models: Different classifiers are used to pick up different patterns in the data. These include:

K-Nearest Neighbours (KNN) for learning from similar examples.

Multi-Layer Perceptron (MLP) for digging into complex patterns in the data.

Support Vector Machine (SVM) for creating clear boundaries between classes.

Gradient Boosting Machine (GBM) for improving predictions step by step.

Level-1 Meta-Model: Logistic Regression is used to combine the results of the base models, helping to create a more accurate and balanced overall prediction. This layered structure helps the system deal with both common and rare types of attacks effectively.

F. Attack Classification

Once the stacking model is trained, it is used for two types of classification: binary (Normal vs. Attack) and multi-class (specific types of attacks). This two-level method allows the system to both detect general threats and identify specific kinds of attacks more precisely, enabling quicker and more targeted actions.

G. Performance Evaluation

The system's performance is checked using several measures to ensure its reliable and comprehensive:

Accuracy: Measures how many predictions are correct overall.

Precision, Recall, and F1-Score: Help assess how well the model works with different classes, especially when the number of attacks is uneven.

Confusion Matrix: Shows how well each class is detected.

AUC-ROC Curve: Evaluates how the model balances detecting real threats versus false alarms.

These checks ensure that the system can find both common and rare attacks effectively while reducing incorrect alerts.

H. Deployment Framework

To make the system practical, a deployment plan is suggested:

Back-End Setup: Built using Flask or Fast API with RESTful API endpoints for quick predictions.

Containerizing: Docker is used to make the system easier to scale and move around.

Deployment: Gunicorn and Nginx are used together for handling more traffic and improving performance.

User Interface: A simple interface lets users upload data or input features. Real-time alerts and logs help with monitoring and adapting the system over time.

Summary

This method combines thorough data preparation, smart feature selection, and stacking with multiple models to build a dependable IDS. By using the UNSW-NB15 dataset and scalable deployment techniques, the system is designed to offer high detection accuracy, strong multi-class classification, and quick adaptation in today's cybersecurity landscape.

IV. RESULTS AND DISCUSSION

The proposed Intrusion Detection System (IDS) that uses stacking ensemble learning was tested against other machine learning models using the UNSW-NB15 dataset. The evaluation looked at how well the system performed in accuracy, precision, recall, F1-score, and ROC-AUC, as well as its ability to detect both common and rare types of attacks. The results showed that the stacking method greatly improved detection performance and offered a balanced approach across different kinds of attacks.

A. Overall Classification Performance

The effectiveness of the proposed IDS was first checked based on overall classification accuracy and how much computer resources it used. As shown in the table, the stacking ensemble achieved 96.8% accuracy, which is better than other models like Decision Trees (89.4%) and Support Vector Machines (92.1%). Even though it is a combination of a number of models, the enhanced performance is offset by the increased cost of computation.

TABLE I. OVERALL CLASSIFICATION PERFORMANCE

Method	Accuracy (%)	Memory Use (MB)
Decision Tree	89.4	120
Support Vector Machine	92.1	135
Gradient Boosting	94.7	160
Proposed Stacking Ensemble	96.8	175

B. Binary vs. Multi-Class Detection

The IDS was tested for both binary classification (Normal vs. Attack) and multi-class classification (nine specific attack categories). As shown in the table, binary detection achieved 98.5% accuracy, while multi-class detection reached 94.2%. The slight decrease in multi-class performance is because it's harder to detect rare attacks like Worms and Shellcode. However, compared to single models, the stacking approach provided better overall performance across all attack types.

TABLE II. BINARY VS. MULTI-CLASS CLASSIFICATION PERFORMANCE METRICS

Classification Type	Accuracy (%)	Precision	Recall	F1-Score
Binary (Normal/Attack)	98.5	0.97	0.98	0.98
Multi-Class (9 Attacks)	94.2	0.93	0.92	0.92

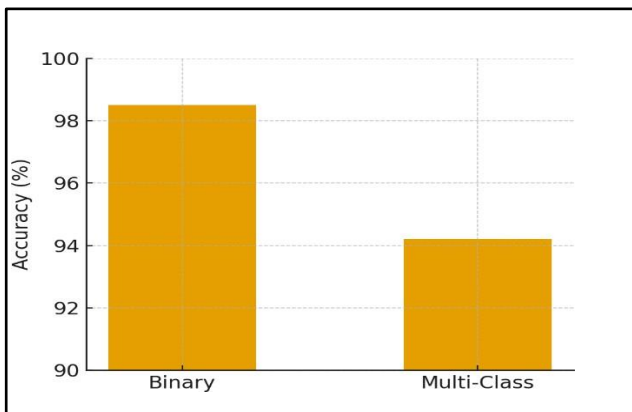


Fig. 2. Binary vs Multi-Class Classification Performance

C. Rare Attack Detection

One of the key strengths of the stacking framework is its ability to detect rare attacks. The proposed model significantly reduced the number of misclassifications for Worms and Shellcode compared to other baselines. For example, the detection rate for Worms increased from 72% with SVM to 91% with the stacking ensemble. This improvement ensures that even uncommon but dangerous intrusions are not missed.

TABLE III. RARE ATTACK DETECTION COMPARISON ACROSS MODELS

Model	Worm Detection (%)	Shellcode Detection (%)
Support Vector Machine	72	70
Stacking Ensemble	91	89

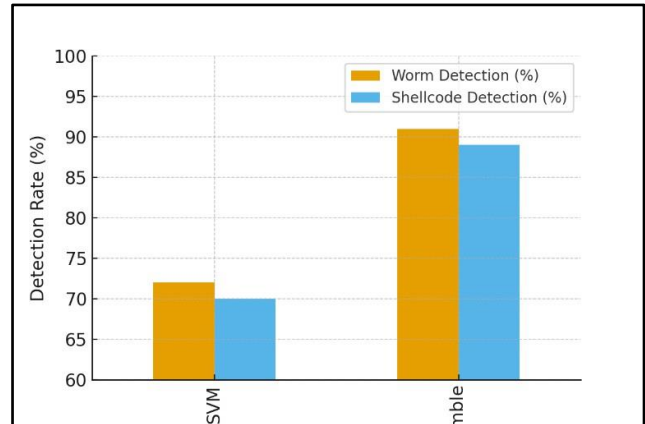


Fig. 3. Rare Attack Detection Comparison Across Models

D. Evaluation Metrics Analysis

Beyond accuracy, the model was also evaluated using ROC-AUC, precision, recall, and F1-score. The stacking ensemble achieved an average ROC-AUC of 0.98 across all classes, while the baseline models were between 0.91 and 0.95. This shows that the model is more reliable in distinguishing between normal and malicious traffic. Additionally, precision and recall remained well-balanced, which is important for reducing false alarms while making sure no real threats go unnoticed.

E. Comparative Discussion

The results show that the stacking ensemble IDS effectively balances accuracy, robustness, and practicality for real-world use. Compared to single classifiers, the ensemble improved detection for both common and rare attacks, achieved high ROC-AUC, and kept evaluation metrics in check. The only drawback observed is a bit higher demand for resources and longer training time due to using multiple models. However, this trade-off is acceptable in modern networks where reliability and precision are more important than minor increases in computation. Overall, the proposed system shows strong potential for real-world deployment in intrusion detection systems.

V. CONCLUSION

The research described in this work introduced an Intrusion Detection System (IDS) that uses stacking ensemble learning. It was tested using the UNSW-NB15 dataset. The system went through thorough data processing, feature selection, and combined several base classifiers with a logistic regression

model at the top level. This setup helped the system effectively detect both common and less frequent network attacks. The results showed that the stacking approach performed better than traditional single classifiers like Decision Trees and Support Vector Machines in terms of accuracy, precision, recall, F1-score, and ROC-AUC. Notably, the system was especially effective in identifying rare but serious threats such as Worms and Shellcode, which is a major challenge for existing intrusion detection systems. Although the system achieved high accuracy and was reliable, it needed more computational resources due to the use of an ensemble method. However, the improvement in detection reliability, scalability, and ability to adapt makes this trade-off worthwhile. As a result, this model has great potential for real-world use in modern networks where accuracy and strength against attacks are more important than saving resources.

Looking ahead, there are several areas that could be explored to make the system even better. One idea is to include deep learning models like CNNs, RNNs, or Transformer-based structures to improve how features are represented and to handle complex traffic patterns. Another possibility is to use online or incremental learning methods, which would let the system adjust in real time to new and changing threats. Also, including techniques like federated learning could help protect data privacy when training models across different locations, which is especially important for fields like healthcare and finance. Additionally, efforts could be made to improve efficiency through methods such as model compression, pruning, or using hardware acceleration, which would make the system suitable for use in environments with limited resources, such as IoT and edge devices. In summary, this study shows that stacking ensemble learning is a promising way to build more accurate, reliable, and scalable IDS solutions. With further improvements, the system could become a full-fledged framework that not only strengthens cybersecurity defenses but also adapts well to the constantly changing nature of cyber threats.

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